THE COZY SCIENCE LAB · ORIGINAL STUDY SERIES

Earth & Space

A Grade 7–10 study handbook with concept notes, examples, self-check explanations and labelled illustrations.

In this handbook

- Plate Tectonics
- Volcanoes & Earthquakes
- Weather, Climate & Typhoons
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Edition 1 · Lab-authored learning materials. Supporting publications are credited with each topic.
Examples are constructed for teaching; diagrams state their model and scale limits. For current Philippine hazard or weather decisions, consult the responsible agencies rather than a static study guide.

Plate Tectonics

Plate tectonics connects patterns in mountains, ocean floors, volcanoes and earthquakes.

Moving solid plates

The lithosphere includes crust and rigid uppermost mantle. Plates move over deeper material that can deform over long times. Earth's mantle is mostly solid; it is not a global ocean of liquid magma.

Boundaries

Divergent boundaries separate, convergent boundaries approach, and transform boundaries slide past. The effects depend on the plate materials and the particular setting; a convergent boundary does not always make the same landform.

Evidence and the Philippines

Seafloor ages, magnetic patterns, earthquakes, rock evidence and modern position measurements support plate motion. The Philippines lies in an active tectonic setting with complex plate interactions and subduction zones.

Key terms

Subduction

One plate descending beneath another at a convergent boundary.

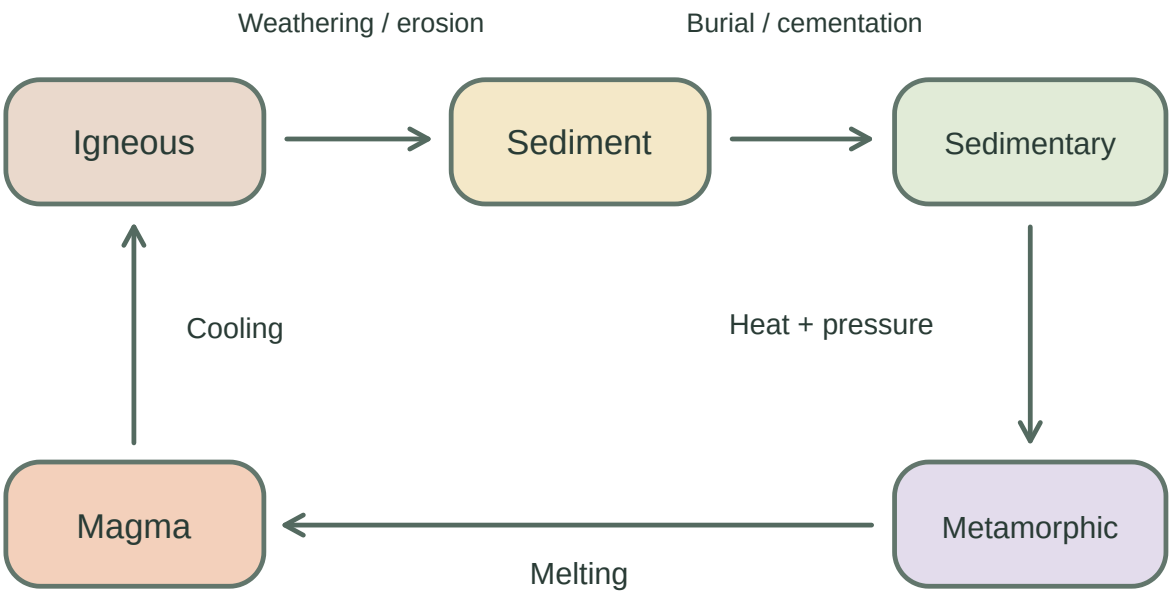
Lithosphere

The rigid outer layer divided into tectonic plates.

Matching several clues

Matching continental outlines can suggest a question. Comparing fossils, rock belts and ocean-floor evidence provides additional, independent tests of plate-motion explanations.

Remember: Plate motion is usually slow on a human timescale, even though an earthquake can release accumulated strain rapidly.



Rocks can take many paths. Selected pathways only. Any exposed rock can weather; several rock types can undergo metamorphism or melting. This is not a compulsory one-way cycle.

Check your thinking

Is the entire mantle liquid because plates move?

Explanation: No. Most mantle material is solid but can deform and flow over geological time.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

U.S. Geological Survey. Earthquake Hazards Program. https://www.usgs.gov/programs/earthquake-hazards

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Volcanoes & Earthquakes

Explore a volcano from its underground magma to its surface vents, then compare volcanic hazards with earthquakes. Both can relate to tectonic activity, but they arise through different processes.

Inside a volcano

Magma is molten or partly molten material below the surface. It can move through passages to a vent, where lava, gases or fragments may emerge. A crater is a surface depression around a vent. Real volcanoes have different shapes and plumbing systems; magma does not come from Earth's outer core.

Lava, ash and dangerous flows

Lava is erupted molten rock. Volcanic ash consists of tiny rock, mineral and glass fragments. Pyroclastic flows are fast, hot mixtures of gas and volcanic material. A lahar mixes water with volcanic debris and can travel down river valleys, including after an eruption.

Why eruptions vary

Magma composition, viscosity, gas content and changing pressure influence eruption behaviour. Gas can become trapped in viscous magma, supporting explosive activity. A volcano's appearance alone cannot tell you whether an area is safe.

Earthquake sources

An earthquake often begins when rocks slip along a fault and release elastic strain energy. The focus is the source within Earth; the epicentre is the point on the surface above it.

Waves and measurements

P waves travel through solids and fluids; S waves do not travel through liquids. Magnitude describes earthquake size using a defined scale, while intensity describes observed effects at a location.

Volcanic systems

Magma is molten or partially molten material below the surface; lava is erupted material at the surface. Eruption behaviour depends on factors including composition, gases and conditions in the system.

Key terms

Magma

Molten or partly molten material below Earth's surface.

Lava

Molten rock after it reaches the surface.

Vent

An opening through which volcanic material reaches the surface.

Crater

A surface depression surrounding a volcanic vent.

Lahar

A flowing mixture of water and volcanic debris.

Pyroclastic flow

A fast-moving, hot mixture of volcanic gas and solid fragments.

Fault

A fracture or zone along which rocks have moved.

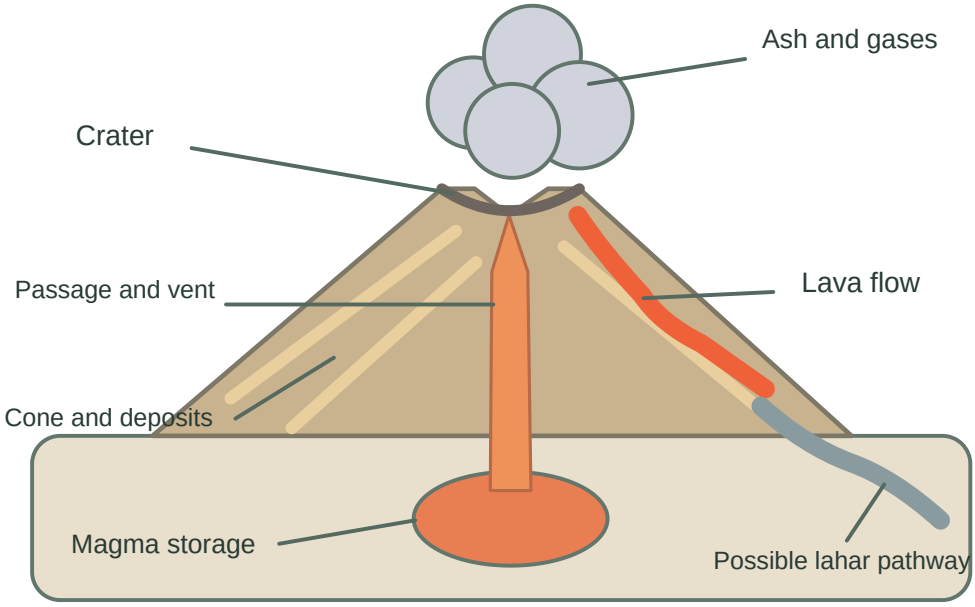
Epicentre

The surface point above an earthquake's focus.

One earthquake, different effects

Two towns can experience different shaking intensity from the same earthquake because distance, local ground conditions and structures differ.

Remember: These are study notes, not a current hazard bulletin or an earthquake prediction. Use current DOST–PHIVOLCS information and your school's preparedness instructions.



A learning model, not a current hazard map

Inside a volcano. Schematic composite-volcano cross-section, not to scale. Real volcanoes vary in shape and plumbing. This diagram is not a hazard map or an eruption forecast; follow current PHIVOLCS and local-authority guidance.

Check your thinking

Does a single earthquake have the same intensity in every town?

Explanation: No. Intensity describes effects at each location and can vary widely.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

DOST–PHIVOLCS. Volcano and Earthquake Information. https://www.phivolcs.dost.gov.ph/

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Weather, Climate & Typhoons

Weather describes atmospheric conditions at a place and time. Climate concerns longer-term patterns and variability.

Air, water and energy

Uneven solar heating, water movement and Earth's rotation help drive atmospheric motion. Water evaporates, condenses and falls as precipitation, linking the atmosphere with land and oceans.

Monsoons and tropical cyclones

Monsoons involve seasonal changes in prevailing winds. Tropical cyclones require a combination of conditions; warm ocean water alone does not guarantee formation. Philippine forecasts and warnings are issued by DOST–PAGASA.

Climate and evidence

Climate conclusions use long records and multiple kinds of evidence. Greenhouse gases absorb and emit infrared radiation, affecting Earth's energy balance. The greenhouse effect and ozone depletion are different topics.

Key terms

Humidity

A measure concerning water vapour in air; the meaning depends on the particular measure used.

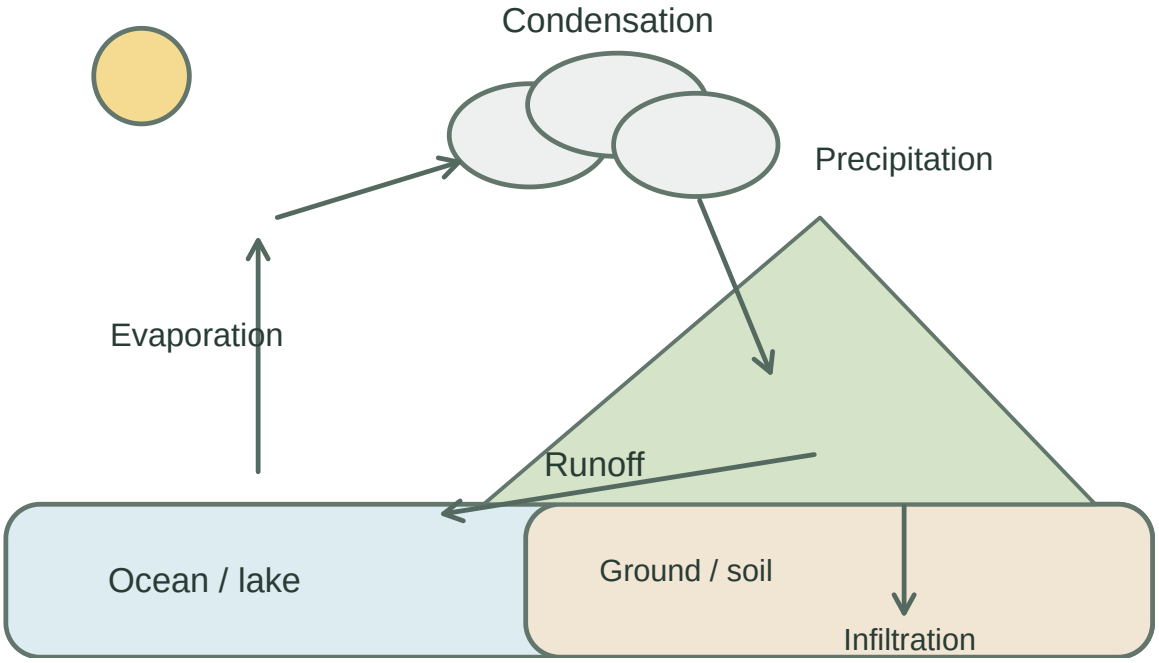
Climate

Long-term statistical patterns of weather and related conditions.

A rainy day

One unusually rainy day describes weather. Comparing rainfall records across many years helps investigate climate patterns and changes.

Remember: A science diagram is not a live weather forecast. Consult current PAGASA advisories for weather decisions.



Follow a water molecule. Simplified pathways, not to scale. Water vapour is invisible; clouds contain tiny liquid droplets and/or ice particles.

Check your thinking

Does one cool afternoon disprove a long-term warming trend?

Explanation: No. A local short-term event and a long-term climate trend describe different scales.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

DOST–PAGASA. Weather and Climate Information. https://www.pagasa.dost.gov.ph/

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Our Solar System

The Solar System contains the Sun, eight planets, dwarf planets, moons and smaller bodies. Use models while remembering their scale limits.

Our place in space

The Sun is a star. Earth is a planet with one natural satellite, the Moon. Pluto is a dwarf planet; the eight recognized planets extend from Mercury to Neptune.

Phases and eclipses

Moon phases result from our changing view of the Moon's sunlit half. An eclipse involves a particular alignment and shadow. Earth's shadow is not the explanation for the ordinary cycle of Moon phases.

Seasons

Earth's axial tilt changes the angle and duration of sunlight over a year. Opposite hemispheres experience opposite seasons. Changing Earth–Sun distance is not the primary cause of seasons.

Key terms

Orbit

A gravitationally influenced path around another body or a shared centre of mass.

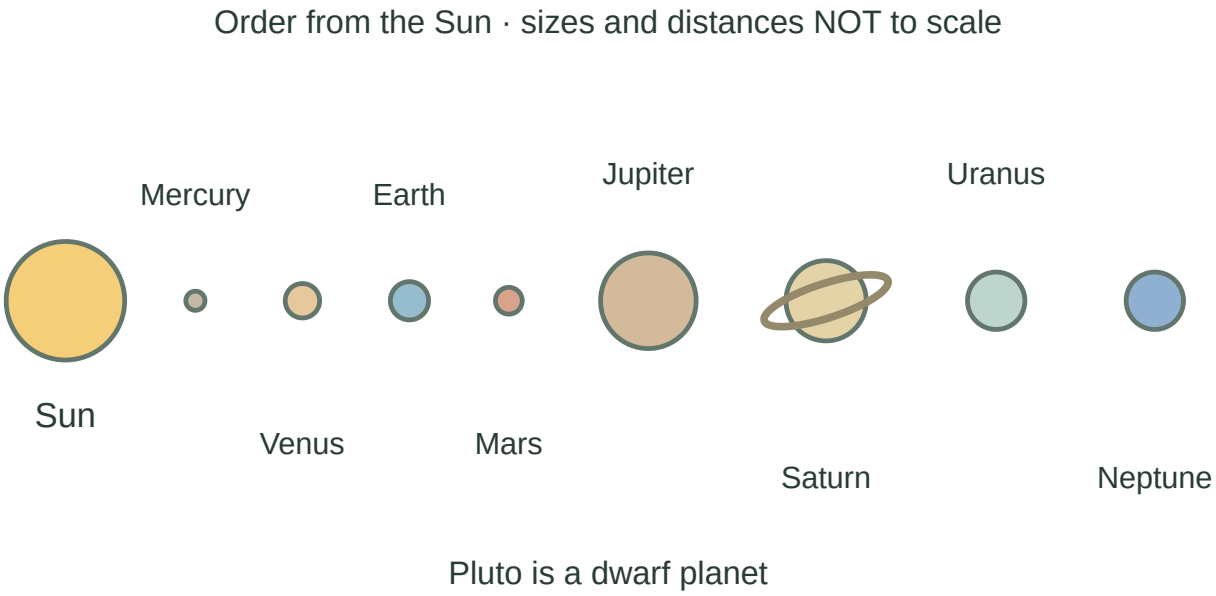
Axial tilt

The inclination of a body's rotation axis relative to its orbital plane's perpendicular.

Opposite seasons

When the Northern Hemisphere is tilted toward the Sun, it receives longer days and more direct sunlight while the Southern Hemisphere is tilted away.

Remember: A classroom Solar System picture cannot show both sizes and distances accurately at the same convenient scale. Read its scale note.



Eight planets, one star. Planet sizes, distances and spacing are deliberately NOT to scale. The Sun is a star; Pluto is a dwarf planet, not a ninth planet.

Check your thinking

Is a half Moon normally caused by Earth's shadow covering half of it?

Explanation: No. We see part of the illuminated half from our viewing direction; Earth's shadow is involved in a lunar eclipse.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

NASA Science. Solar System Exploration. https://science.nasa.gov/solar-system/

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Stars & Galaxies

Astronomy studies very large distances and long histories using light, physical models and observations.

Stars and energy

Stars such as the Sun produce energy through nuclear fusion in their interiors. Their surface temperatures relate to colour; hotter stellar surfaces tend to appear bluer than cooler red ones.

Brightness and distance

Apparent brightness depends on both a source's luminosity and its distance, as well as intervening material. A light-year is a distance: the distance light travels in one year in a vacuum.

Galaxies and scale

Our Sun is one star in the Milky Way galaxy. Galaxies contain stars, gas, dust and other matter. Stellar life histories depend strongly on mass, so not every star ends its life in the same way.

Key terms

Luminosity

The rate at which a source emits energy.

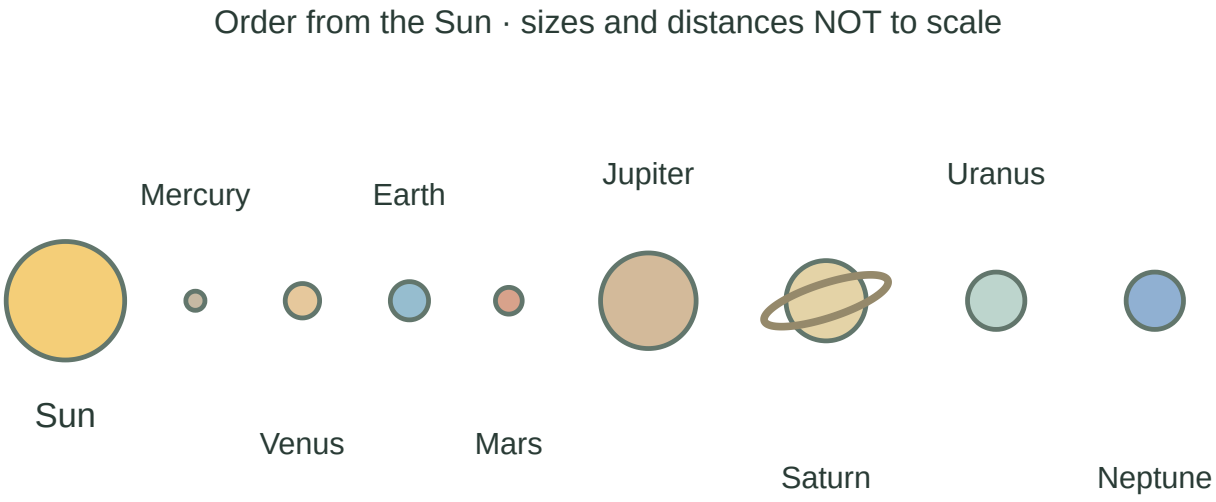
Light-year

A unit of distance, not an age.

Seeing the past

Ignoring other complications, light from a star 10 light-years away takes about 10 years to reach us. We observe information emitted years earlier.

Remember: A star that looks dim is not necessarily intrinsically faint. It may be much farther away.



Eight planets, one star. Planet sizes, distances and spacing are deliberately NOT to scale. The Sun is a star; Pluto is a dwarf planet, not a ninth planet.

Check your thinking

Why is 'the star is five light-years old' a unit mistake?

Explanation: Light-years measure distance. An age needs a time unit, such as years.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

OpenStax · Rice University. Astronomy 2e. https://openstax.org/details/books/astronomy-2e

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Rocks & the Water Cycle

Rock and water cycles describe connected processes. Neither is a single compulsory route followed by every sample.

Rock formation

Igneous rocks form when molten material cools and solidifies. Sedimentary rocks can form as sediments are deposited, compacted and cemented. Metamorphic rocks form through changes caused by conditions such as heat and pressure without complete melting.

Changing landscapes

Weathering breaks down or alters material. Erosion transports it, and deposition places it elsewhere. Many different pathways connect rock types over geological time.

Water in motion

Solar energy supports evaporation and gravity helps move water through precipitation, runoff and groundwater flow. Plants release water through transpiration. Groundwater often fills pores and fractures rather than giant open underground lakes.

Key terms

Weathering

Breaking down or chemically altering material near Earth's surface.

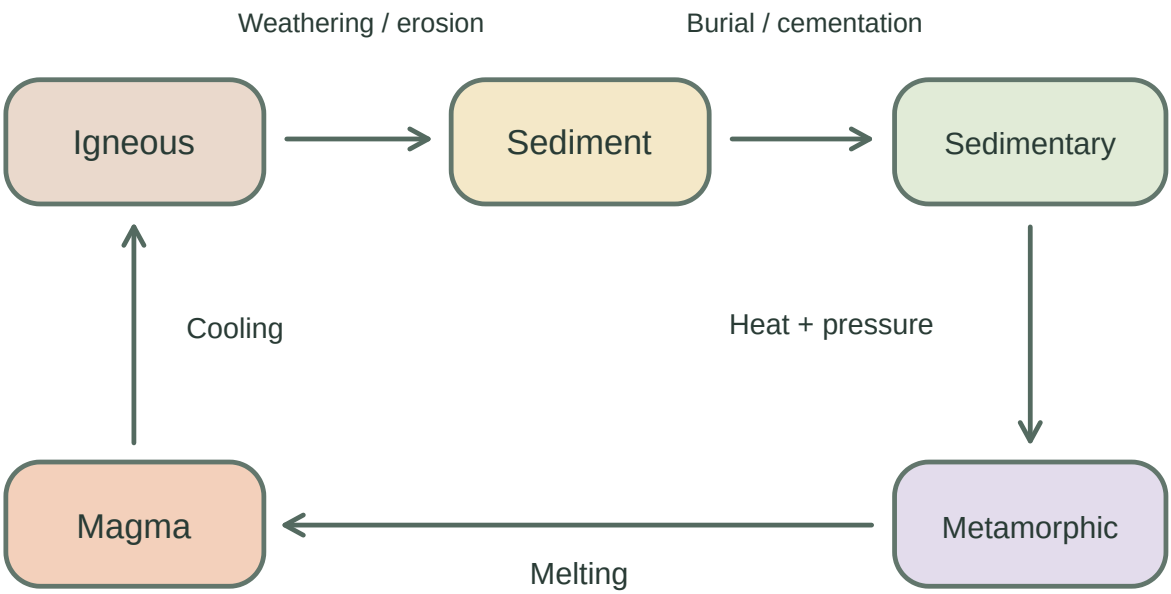
Infiltration

Water entering the ground from the surface.

A grain of sand

Weathering can release a grain from a rock. Water can erode and transport it; slowing water may deposit it. Burial and other processes can later help form sedimentary rock.

Remember: Metamorphism is not the same as melting. Once rock melts and then solidifies, the resulting rock is igneous.



Rocks can take many paths. Selected pathways only. Any exposed rock can weather; several rock types can undergo metamorphism or melting. This is not a compulsory one-way cycle.

Check your thinking

Can an igneous rock become sediment without first becoming metamorphic?

Explanation: Yes. Weathering and erosion can produce sediment from any exposed rock type under suitable conditions.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

U.S. Geological Survey. Water Science School. https://www.usgs.gov/special-topics/water-science-school

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf